DIGITAL IMAGE PROCESSING

UNIT 1

2 MARKS

1.What is a Digital image?

Ans. Digital Image is a multidimensional function. It is represented as f(x, y, z, λ) where x and y are spatial (plane) coordinates. f(x, y, z, λ) represents three-dimensional (3-D) color image, whereas f(x, y, λ) represents two dimensional (2-D) "color image" and f(x, y) represents the "grayscale image"**.**

2.What is a Digital image processing?

Ans. Digital Image Processing (DIP) is the use of computer algorithms to perform operations on digital images to enhance them or to extract useful information. It involves converting images into digital form and applying techniques such as filtering, enhancement, restoration, compression, and analysis to improve image quality or obtain meaningful data.

3.List the any four steps of Digital Image Processing ?

Ans. Image Acquisition

Image Enhancement

Segmentation

Compression

4.What is Morphological Processing and Segmentation?

Ans. **Morphological Processing:** Morphological processing involves tools for extracting image components useful for representing and describing shape.

Segmentation: Segmentation procedures partition an image into its constituent parts or objects

5. What is Image Enhancement and Image Restoration?

Ans. Image Enhancement: Image enhancement is among the simplest and most appealing areas of digital image processing, such as, changing brightness & contrast etc.

Image Restoration: Image restoration is an area that also deals with improving the appearance of an image.

6. Define grayscale image. How are gray-scale images represented?

Ans. A gray-scale image is a data matrix whose values represent shades of gray without any apparent color

The darkest possible shade is black, which is the total absence of transmitted or reflected light. The lightest possible shade is white, the total transmission or reflection of light at all visible wavelength s.

7.What is an indexed image? Write its components.

Ans. An indexed image is a digital image where pixel values are indices (pointers) to a color table

- data matrix of integers X
- Color map matrix map.

8. What is image matrix representation?

- An image matrix representation means storing a digital image as a 2D matrix of pixel values.

- Each element of the matrix corresponds to one pixel in the image.
- The value of each element represents either intensity (grayscale) or color components (RGB).

9. What is pixel intensity?

Ans. Pixel intensity is the numerical value that represents the brightness or color strength of a pixel in a digital image. It tells us how light or dark that pixel appears.

10. Define Scotopic, Photopic and Mesopic vision

Ans. Scotopic Vision

- Vision under very low light (night vision).
- No color perception, only shades of gray.
- Highly sensitive to light but poor in detail.

Photopic Vision

- Vision under **bright light (daylight vision)**.
- Provides **color perception** and sharp details.
- Less sensitive to faint light compared to rods.

Mesopic Vision

- Vision under **intermediate light levels** (dawn, dusk, street lighting).
- Partial color perception with moderate detail.
- Transition zone between scotopic and photopic vision.

11. What is brightness adaptation?

Ans. Brightness adaptation is how our eyes adjust to different light levels. This ability allows us to see clearly in both dark and bright environments by changing how sensitive our eyes are to light

12. Define intensity discrimination.

Ans. In Digital Image Processing (DIP), intensity discrimination is the ability to tell apart different brightness levels of pixels. It measures how small a difference in brightness can be noticed or shown.

13. Define Sampling and Quantization.

Ans. Sampling

- Sampling is the process of converting the spatial coordinates (x, y) of an image into discrete values.
- It divides the image into a grid of pixels.

Quantization

- Quantization is the process of converting the **amplitude (intensity) values** of pixels into **discrete levels**.
- Each pixel's brightness is mapped to a finite set of values.

14. What is RGB image representation?

Ans. RGB Image Representation is a method in digital image processing where each pixel is defined by three components

— Red, Green, and Blue intensities. These three values combine to produce a wide range of colors, making RGB the standard model for representing and displaying color images**.**

15. What is HSV color model?

Ans. The HSV colour model describes colours using three values: Hue (colour type), Saturation (colour strength), and Value (brightness). It represents colours in a form closer to how humans naturally see and understand them.

16. What is YCbCr model?

Ans. YCbCr Model → A color model that represents an image using Y (luminance/brightness), Cb (blue-difference chroma), and Cr (red-difference chroma).

17. What is image cropping?

Ans. The process of cutting out unwanted outer parts of an image to keep only the desired region.

18. What is image resizing?

Ans. Changing the width and height of an image, either enlarging or reducing it.

19. What is image complementing?

Ans. Creating the inverse of an image by replacing each pixel value with its opposite (black ↔ white, bright ↔ dark).

20. Define 4-neighbors and 8-neighbors of a pixel.

Ans. 4-Neighbors: Pixels directly above, below, left, and right of a given pixel.

8-Neighbors: Pixels in all directions around a given pixel (including diagonals).

21. What is adjacency and connectivity?

Ans. Adjacency → Two pixels are adjacent if they share a common boundary (depending on 4-, 8-, or diagonal neighborhood).

Connectivity → A set of pixels is connected if they are adjacent to each other and belong to the same region based on intensity or property**.**

22. List any four applications of Digital Image Processing.

- Medical Imaging → Used in X-ray, MRI, CT scans for diagnosis.
- Remote Sensing → Satellite images for agriculture, weather, environment.
- Industrial Inspection → Detecting defects in products automatically.
- Security & Surveillance → Face recognition, motion detection, CCTV monitoring